

Total No. of Questions :10]

SEAT No. :

P3270

[5670]-539

[Total No. of Pages :3

B.E. (Mechanical Engineering)

ENERGY AUDIT & MANAGEMENT

(2015 Pattern) (Semester-I) (Elective-II) (402045C) (End Sem.)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Draw neat diagrams wherever necessary.
- 3) Figures in the right indicate full marks.
- 4) Use of Scientific Calculator is allowed.
- 5) Assume suitable data whenever necessary.

- Q1)** a) Discuss environmental impacts due to conventional energy use. [5]
b) Write short note on Energy Audit Software. [5]

OR

- Q2)** a) Write a short note on Energy Policy. [5]
b) Explain Energy Audit Methodology in Ice Factory. [5]

- Q3)** a) A typical manufacturing facility has a 200JP compressor (which requires 215 bhp) that operates for 7200 hrs annually. It is fully loaded 85% of the time (motor efficiency=95%) & unloaded the rest of the time(25% full-load bhp & motor efficiency=90%). The aggregate electric rate is Rs. 3.5/kWh. Calculate COE. [5]
b) Consider 2 projects, A&B with the initial investment Rs.80000. The project life times are 5 years in case. The saving in each of the 5 years for the two projects are as shown in following table ROI = 20%. [5]

	Project A	Project B
Capital Cost (Rs.)	Rs.80000	Rs.80000
Year	Cash flow (Rs.)	Cash flow (Rs.)
1	10000	50000
2	20000	40000
3	30000	30000
4	40000	20000
5	50000	10000

OR

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- Q4) a)** Calculate internal rate of return for the project which has following cash flow, discount rate is 10%. [6]

Investment	Rs. 10,00,000
Saving in year	Cash flow (Rs.)
1	2,00,000
2	2,00,000
3	3,00,000
4	3,00,000
5	3,50,000

- b) A company borrows Rs. 30000 to finance a new boiler installation. If the interest rate is 12% per annum & repayment period is 6 years. Calculate the value of total repayment & monthly repayment value for [4]
- i) Simple Interest ii) Compound interest

- Q5) a)** What are the different heat losses occurring in oil fired furnace? Explain in brief. [8]

- b) A centrifugal pump is pumping $85 \text{ m}^3/\text{hr}$ of water and pressure rise in the pump is 6 kg/cm^2 . If power drawn by motor is 25 KW . Find out the pump efficiency. Assume motor efficiency as 90% & water density as 998 Kg/m^3 . [10]

OR

- Q6) a)** Explain the energy saving opportunities in compressed air system. [8]

- b) Calculate pump efficiency from the data given: pump flow is $0.40 \text{ m}^3/\text{s}$, power absorbed: 325 KW , suction head + 1 m , Delivery head 55 m , motor efficiency 88%, type of drive: direct coupled, density of water 996 kg/m^3 . [10]

- Q7) a)** The connected load for the hostel are as below.

- i) 190 Fluorescent tubes of 55 W each with magnetic ballast.
 ii) 20 Fluorescent tubes of 40 W each with electric ballast.
 iii) 20 old fan of 100 W each.

It is decided to replace the all tubes of 20 W 7 Old fan by a new fan of 80 W . Considering usage of 6Hrs per day & an electrical tariff of Rs. 4 per KWh. Calculate energy saving of tubes & fans replacement. [8]

- b) What are the factors affecting energy efficiency of electrical motor? [8]

OR

